MSX and SPECTRAVIDEO

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MSX and SPECTRAVIDEO

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EDITORIAL

First off I think I owe you all an apology for our magazine being so late, we are really trying to make a go of this magazine and to keep it alive for all MSX & SPECTRAVIDEO users.

Please bare with us I think we have almost got the hang of doing 8-10 hours a day at work and then another 4-6 hours at home, then all day saturday, mind you we do have Sunday's off.

One of the biggest reasons or excuse that we have is that after - Vol:1 No:12 their were a lot of people up for re-subscription and we were a bit worried that we would lose subscribers in our take-over,
Well we did, we lost more than 100 subscriptions (142) to be exact from reasons of Selling their computers to moving overseas.

So please if you know someone not subscribing help us to help you keep this magazine going.
If you feel let down because the magazine is late, just imagine how we feel after losing the support of all those people.

We have received Input from various people and all of it is from Australia what about our **overseas** readers dropping us a line or 3,
I think there is a lot of people who would like to hear from you, let us know what you know about MSX or SVI.

FRONT COVER WINNER

This months front cover winner is ---- KAYE BERGMAN of Queensland
Kaye is only 8 years old so I hope this inspires other people.

We have some news which is good for us but not so good for all the people who buy MSX COMPUTING.
Unfortunately they have ceased to exist due to high taxes in England which is a real blow for them but can be quite good for us, if we can get into the English market our software dealers could buy software and hardware at reasonable prices which they can then sell to us at a "Reasonable price".

We will also endeavor to get people from England and surrounding country's to subscribe to us this will give us more advertising, as well as a bigger magazine.

Also this could be good for SVI users because there must be more people in this world that use SV machines apart from the ones that subscribe to Computer Forum.

More good news !!!.. How much can you take in one reading session ?? We are trying to push MSX2 sales, in Australia at the moment it is being sold by SONY & PHILLIPS but still is not being advertised.

We received a letter from the ASCII corporation and they asked us if we could find dealers who would support MSX2 in Australia, and prior to this addition going to print, we sent them a list of 5 Dealers which they could go through and hopefully send out the various information required by for our dealers, now all we can do is hope our dealers like ASCII's deal and if so we can then have the opportunity to buy MSX2 machines and for those of you who have not seen any MSX2 machines at all they are a superb looking, as well as acting machine.

As far as running your old software on an MSX2 machine, and I quote from a small orange furry fella "NO PROBLEMS", well that is what I was told. As far as new software goes I should imagine that with the push of MSX2 it will be in ample quantity, mind you MSX2 has been out in most European countries for years, thus building up a massive software collection that we have never even heard of.

I'm a bit disappointed to the response we got to our new ideas we only had a few people that actually wrote into us to start up our new columns, so if at all possible what about putting pen to paper and sending in that HIGH SCORE, SOFT-ADD, PEN PAL or ADVENTURE query.

If you really look at our magazine you will see that 60% of it would be input from our readers 30% is information to our readers, and the other 10% is split up between advertising (which is very small) and just plain waffle to fill up space.

So please compare our magazine with another computer magazines and see if their's could match our percentages, and I hope this tells you that we rely on reader input to exist, we are not asking for 10 pages a month, 9 would be plenty.

I leave you in thought... Bye till next time.

KNOW-IT-ALL

The following letter was recieved from Chris McRobb of W.A. and is addressed to ALL of US that think we know it all, from one who found out, the hard way, that he didn't.

Take it away Chris ....

How many out there have ever read there manuals from the start? Not many I bet.

The first few pages are usually trademark and copyright acknowledgments, after that the user cautions, these are very important, if they weren't they wouldn't bother puting them there.

There are several very important rules to follow...

1: Power Up; in this order disk drive, computer,printer etc.

2: Power Down; printer etc, remove disk from drive, computer, then disk drive.

3: NEVER remove the disk while the "IN USE" lamp is on, or you may corrupt your disk.

4: NEVER remove a cartridge with the power on, doing so will cause amnesia, and a computer with no memory is quite boring.

5: Keep ALL liquids well away from the computer and peripherals. Liquids and electronics don't mix.

6: Avoid using your computer during thunderstorms, a bolt of lightning down the line can fry your computers internals and in extreme cases fry the user as well, also remove the plugs from the wall as the switch on your computer will not stop the high voltage jumping into the computer.

Did you know that the reset button only resets basic programs and not machine code? So to remove machine code programs you must turn the computer off for at least 5 seconds to allow discharge.

Don't forget, Imagination is more important than knowledge. So imagine what could go wrong, and it probably will.

CHRIS

CLUB SCENE

Here is the first of, we hope, many pages of news and helpfull hints from User Groups and Clubs around Australia.

The first is from: MSX CLUB (AUSTRALIA)
C/- ROGER STOCKBURGER,
P.O. BOX 443,
RICHMOND, N.S.W. 2753

A lot of Software is supplied on cassette. Fine if you are not priveleged with the presence of a Diskdrive occupying one of your cartridge slots.

Fellow disk users, we can back up SOME cassette software to disk. All that is needed is the START, END and EXECUTION addresses of the machine code file(s) that make up your favourite game. Many of you don't have the software to find this data. That is why we have written into this column and any other clubs or "Hackers" who have anything to add, please write in and support **OUR** magazine.

VESTRON: (Players)
This program consists of a single block of machine code and is run using the command: BLOAD"CAS:",R Backing up this program may be tricky, as it occupies some disk area.

Start Address: &HA440
End Address: &HF230
Exec. Address: &HA440

VAMPIRE: (Code Masters)
Four programs make up this game... A Basic loader followed by three machine code programs.

First, load the basic program with the LOAD"CAS:" Command and then save it to disk using SAVE"VAMPIRE.BAS".

SCREEN (M/CODE)
Start Address: &H8598
End Address: &HA300
Exec. Address: &H8598

MAP1 (M/CODE)
Start Address: &H8598
End Address: &HC1f2
Exec. Address: &H8598

As this program doesn't use disk area as it is, back-up to disk is straight forward. But, before we start, ensure that the basic loader program is suitably changed as follows.

GAME (M/CODE)
Start Address: &H8598
End Address: &HDB00
Exec. Address: &h8598

Modify these lines as shown:-

```
20 BLOAD"VAMPIR1.BIN",R
21 BLOAD"VAMPIR2.BIN",R
22 BLOAD"VAMPIR3.BIN",R
```

Now save this program back to disk as "VAMPIRE.BAS" .

Let's load the three machine code files and save them to disk in succesion:

Machine Code File #1:- BLOAD"CAS:" <CR>
and when loaded:- BSAVE"VAMPIR1.BIN",&H8598,&HA300,&H8598 <CR>

Machine Code File #2:- BLOAD"CAS:" <CR>
and when loaded:- BSAVE"VAMPIR2.BIN",&H8598,&HC1F2,&H8598 <CR>

Machine Code File #3:- BLOAD"CAS:" <CR>
and when loaded:- BSAVE"VAMPIR2.BIN",&H8598,&HDB00,&H8598 <CR>

Time well spent! You have backed up Vampire to disk. Now, to run VAMPIRE from disk, just enter...

RUN"VAMPIRE.BAS"

What? You haven't got these two programs? Both Vampire and Vestron are available in Australia through MSX SOFTWARE IMPORTS (See the games review section for there address. ED.) at around $10.00 each.

Numerous games supplied on cassette cannot be readily backed up to disk, as they often use thier own loaders (called Headerless Loaders) to prevent copying of the program (and reduce Piracy).

This trend of protecting software is already on the decrease in software from the U.S.A. - such as the U.S. Gold range of games (Grog's Revenge, Spy V Spy, Valkyr, Super Cycle, etc.).

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Welcome to ADVENTURES ANONYMOUS, we kick off this month with an old favorite "THE SECRET OF BASTOW MANOR."

As the story goes,
You are standing on the misty moors outside the walls of BASTOW MANOR old man bastow has dissapeared leaving rumours of his great wealth hidden somewhere in the manor. The baying of the hounds from behind the walls, brings to mind tales of wandering ghosts and hidden traps. Can you find the the treasure solve the mystery .....and live???

Well you most certainly can find the treasure, solve the mystery, but live! Well, that's doubtful.
But you do get eight moves before the dogs attack, if that's any consolation.
Standard commands are used eg. N,S,E and W, LOOK, GO, GET, OPEN, CLIMB, BREAK, MOVE, CUT etc. The graphics are good, I prefer graphics adventures to text adventures at least you can see what you are doing and that can be very useful sometimes. Another feature that can come in handy, is the directions you can go in. A lot of adventures don't tell you the possible exits. Which means you can have a choice of up to ten exits you can choose from including up and down.
Some adventure games boast of up to 200 screen locations. Multiply that by 10 directions and you could be getting nowhere slowly.

As a lot of adventure games and this one is no exception, you have to find several keys to open the doors or grates to the next stage.
Whatever you find should be taken with you, don't leave anything behind. All the items you come across will be used at least once, in some cases twice.

You start out the manor gate, try LOOKing at the gate, you will find what you need to open it. To be able to get it you you will have to go SOUTH and CLIMB.
Don't forget to check out the mailbox, and LOOK twice.
Once inside the gates and inside the grounds, the dogs are upon you, try heading in a SOUTHERLY direction where you will find shelter from the howling hounds.
This is the way into the manor, and LOOK outside the shed you will find a ladder, this will be used for your exit from the manor.
So you could leave it by the ivy, which is 2 EASTS from where you enter the grounds, climbing the ladder will stop the dogs from attacking you, or at least for 8 moves. This is the only exit from the manor and you will need to make another trip outside, for an object that dissapears before you can complete the adventure.
To be able to get inside the manor, you need to get something from the toolbox, this will remove any obstructions from the entrance to the manor. The toolbox is SOUTH a couple of times, from the trapdoor over the snake pit. You will have to drop the branch to be able to cross the pit. Then you return to the door with the beam and remove it.
Once you enter the door you can not return this way, so be sure to take everything with you.

Once inside you are in the north hall. Check out the clock, this will help you get the key from the apple unless you've eaten it.

A word of caution regarding old man bastow. He appears to get extreme pleasure and gratification, going around bopping and dropping things on you. Protection can be found in the cellar SOUTH of the kitchen. Take the meat with you, to get rid of the hounds, But be sure to leave the meat to last. You can only use it once, and you will need it to make your final escape when you have the treasure.

Clues to the combination of the treasure chest can be found in 4 different locations. In the mailbox outside the manor wall, in the library, EAST of the foyer, which is NORTH of the north hall. In the box by the trapdoor, you need a key to open it.
And finally in the bathroom upstairs from the south hall.

If you have all the clues to the treasure chest then it only leaves finding it. Access is from the study in the south hall. The silver key will open the door to the study. Try moving a few things around on the desk, which will open a panel on the wall and will reveal a wheel which you turn. Then make your way back down the trapdoor to the stone cell at the end of the passages. The treasure room door will now be open, and the chest ready to be opened. If you have read all the clues and understand them, then you can now open the chest. If you have not, then you must remember that it is a combination lock and that the letters of the alphabet go around the lock, with the letter A in the topmost spot.
The numbers which you found in the bathroom and in the box should be added together for the first turn, which is to the right.
Then follow the instructions you were given for the next three turns. The chest should open. On leaving the dogs return, just throw the meat and they leave.

A few letters have been received regarding another DOTSOFT adventure, KING SOLOMANS MINES.
The first is from NEIL TURNER from South Australia.

PROBLEM

Neil has a problem getting past the quicksand at the beginning.

ANSWER

You must first THROW the rope, which catches on the tree, and then PULL ROPE 3 times which will take you to the other side. Once across you will not be able to go back. Mind you if you have already picked up the spear and the machette, which were by the dead adventurer you will not need to go back.

The next letter is from PETER ROBINSON from Western Australia.

PROBLEM

Peter can't get across the bridge over the river.

ANSWER

The simple answer to that problem is you don't cross the river by the way of the bridge. But the bridge needs to be down for you to to cross.
First go EAST on bridge then RUN. This will destroy the bridge and leave you back on the same side of the ravine. You must then GO RAVINE , which will take you to the river, which you can cross, and then climb the broken bridge.

PROBLEM
MIKE HERN of South Australia can't open the box he found in the cave.
ANSWER
To open the box you will need the pebbles which are by the river, then you SMASH BOX. There is a diary in the box but I did not find it of any use, in fact just the opposite, after I opened it I could not get past the native he killed me each time. So don't open the box give it to the native along with the necklace which is on the dead zulu, and the native will let you past.

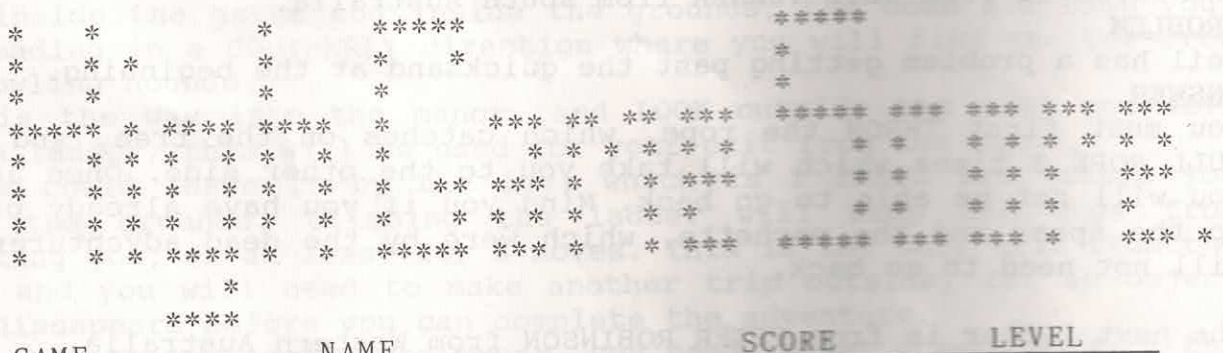
LORDS OF TIME

PROBLEM
ADRIAN WALLACE of New South Wales has a problem with LORDS OF TIME, he can't get into the castle. There is a few things you will have to do. Firstly you should have a gold nugget which you should have gotten from the cavewomen in the STONE AGE for saving her life. You give this to the bartender in the alehouse NORTH of the village green. He will give you ale, which you then take to the messenger, EAST of the cobble square who will in exchange give you some food, don't eat it, take it to the dragon who is in the forrest NORTHISH from cobble square. He will take you to the castle, after you accept his offer.

Please write in if you have any problems,answers,query's or just for a good old fashioned chat and I will endeavor to help you out if required.
Also if you have had problems and been stuck in an adventure, but have solved it, please let me know and I will pass your information on to other adventurers who may be stuck in the same place.
It's really surprising how many people do get stuck in the same place.

Well that's about it for this issue, hope to here from you soon.

bye for now ALAN CRAWFORD.....



GAME	NAME	SCORE	LEVEL
BEAM RIDER	RICHARD JOHNSON	81,112	
BUCK ROGERS	B. RANKIN	122,112	
GOLF	DAVID CLARKE	EVEN PAR	
HYPER SPORTS	P. QUINN	475,000	
LE-MANS	LEE BERGMAN	17,100	
LODE RUNNER	LEE BERGMAN	31,150	
MANIC MINER	KEVIN TATE	66,663	
PIN BALL	B. RANKIN	870,450	
POLAR STAR	R. DUNCAN	210,000	
RIVER RAID	D. OLSON	56,650	

MAILBOX



We received a letter from our good friend Tony James of Victoria and tony asks-

PROBLEM

My problem is the cassette player is giving me a lot of I/O errors presumably because the head is out of adjustment.
A couple of friends who have commodores run a small program that tells them if the head needs aligning (You would have to align your head if you have a comadoor).

As the Spectravideo has the same adjusting screw, I assume that this can be done here, too.

ANSWER

Well I am lucky so far as that my tape deck which by the way is a Spectravideo is used about as much as "a fan in Alaska" so I have not run into that problem, but like tony I would like to know if, when and how do we align the head.
So if any of our readers have had this problem and solved it please let us know.

Our second letter is from Phillip Christie of Victoria and he writes-

PROBLEM

Having recently purchased a copy of "ZEN" Assembler I have noticed a serious bug.
After typing in a program and then attempting to save it to tape as an ASCII file by using the inbuilt "W" command, the file appears to write to tape but when you try to verify or read the so-called file, it does not verify or load in a usable form.
I would like to know if any of our readers have encountered the same problem and if there is a way of modifying the program to make it work properly.

I have a Spectravideo 728 MSX computer and a SVI 767 Data cassette.

ANSWER

Hmmmmmmmmm interesting problem, I have used ZEN but I have not used it in that way, but now that you have pointed out the bug I shall have a go at solving it, and if any of our readers have already solved it please let us know.

Here is a letter from D.M. Stewart of Queensland and he writes-

PROBLEM

I would like to ask about an error which has just recently occurred to me. It is the 'DISK ERROR' and it has happened twice, the first was when I typed in "TITAN LAND" from the magazine MSX COMPUTING and the second time was when I tried to save a machine code program between address 59200 and 59353.
When I tried to BSAVE the program at this address the computer RESET itself and when I tried to call the disk for the program I got a 'DISK ERROR' and even the files command didn't work, the disk was useless.
The game out of MSX COMPUTING has a lot of machine code data statements at the end of it.

ANSWER

By the sounds of what you did and the address that you called you are trying to save the program in the disk operating area which is a definate no no as far as the disk drive is concerned, and that is why it resets itself.
So unfortunately it is a tape based program.

Here is a letter from Justin Freeman of the A.C.T. and he writes-

PROBLEM

1.. I have a program that I typed in from my Toshiba manual which dumps pictures to the printer unfortunately they come out very small can you help out.

2.. I read in another magazine that the Sanyo Light-Pen doesn't work with the Toshiba HX10 is this true?

ANSWER

1.. Without actually seeing the program I would say it dumps out the same length & width as the screen which depending on your default sizes which is normally 39 x 21.
But I don't know how to enlarge the pictures, does any of our readers?

2.. Yes it is true that the Sanyo Light-Pen does not work on the Toshiba HX10, which was tried at the MSX user's group meeting in Western Australia.

Now that you have tapped all my knowledge and I have shared my vast wealth of information with you I bet you think -
Does this bloke know anything or is this the 10% waffle.???

HACKERS CORNER

1. CP/M column
2. Pipeline
3. Suggestions ?
4. MSX - SV compatibility
5. How to write games, my method

BY

ROBERT BRINKWORTH

1..

CP/M COLUMN

As discussed in the last issue a very good symbolic dis-assembler called DASMZLG is available, this version of the program is a Z80 dis-assembler while it's predecessor was called DASM and was for the 8000 uP. Anyway, to use this program for dis-assembling a game that resides in Disk Basic between 8800H and C800, first it must be block moved down to 0100H.

This can be achieved by a simple machine code program written as follows:

```
start: ld    a,0FH     ;switch to 64k RAM
       out   (88H),a
       ld    a,0DFH
       out   (8CH),a
       ld    hl,8800H  ;start address
       ld    de,100H   ;destination address
       ld    bc,0400H  ;length
       ldir            ;block move data
       ld    a,0FH     ;switch ROM back in
       out   (88H),a
       ld    a,0FFH
       out   (8CH),a
       ret             ;return back to caller
```

To run this in BASIC do this:

```
10  DATA 3E,0F,D3,88,3E,DF,D3,8C
20  DATA 21,00,88,11,00,01,01,00
30  DATA 04,ED,B0,3E,0F,D3,88,3E
40  DATA FF,D3,8C,C9,XX
50  AD=&H8200
60  READ D$:IF D$="XX" THEN 100
70  POKE AD,VAL("&H"+D$)
80  AD=AD+1:GOTO 60
100 DEFUSR0=&H8200
110 A=USR0(0)
120 END
```

When this has been run then you can use CP/M. Insert your CP/M disk into your disk drive and close the drive door, if you have a reset switch installed press it, otherwise type in DEFUSR=0:A=USR(0) and the computer will reboot and load up CP/M giving you the A> prompt when it is ready for input.

Now before you enter anything type SAVE 40 (filename) where the word filename is the name of the game or utility that you have block moved, this will save a memory image of the file in memory. After you have completed this menial task, then type DASMZLG and press ENTER.

Leaving you in suspense I will return next time to another exciting section of...CP/M COLUMN

---

2. **PIPELINE**

Thanks to an overwhelming response (over 5 people !!!) to my plea in the last issue about reader response I have decided that at least some people want some machine code programs. Therefore I will try to finish my proposed programs.

---

3. **SUGGESTIONS**

If you have any machine code relevant information or would like anything known then please write, I would only be to happy to give it my best shot at completing what you have indicated for me to do.

---

4. **MSX & SPECTRAVIDEO COMPATIBILITY**

The difference between these computers is like chalk and cheese, but which one is chalk and which one is cheese as they both have slightly different aspects about them which appeal to me.

One thing that towers over the SPECTRAVIDEO is MSX BIOS calls, while the SPECTRAVIDEO has BIOS calls they are not as extensive as the MSX and no information is available for them.

Comparing:

Aspect	MSX	SPECTRAVIDEO
CPU	Z80A	Z80A
SOUND CHIP	8910	8910
VIDEO CHIP	9919	9918
PIA CHIP	8255	8255
CARTRIDGE PORT	YES	YES
EXPANSION PORT	YES	YES
LPT INTERFACE	YES	NO
RESET SWITCH	YES	NO
J/STICK TRIGGERS	2	1
32k,48k,64k SELECTABLE RAM	YES	32k,64k only

Note that there is a very close similarity as the fact is that the SPECTRAVIDEO was actually the first MSX standard computer, but then as life generally goes, MICROSOFT changed the standard and SPECTRAVIDEO didn't stop making the 318 & 328 series until they released there own proper MSX standard computer the 728 & 738.

---

5. HOW TO WRITE GAMES - MY METHOD

The technique I use when writing games is this:

1. Design your screen layouts.
2. Decide what screen mode you will use.
3. Work out the graphics patterns.
4. Work out the sprite patterns.
5. Work out the sound processing.
6. Work out step by step the algorithms needed for your game.
7. Build up a basic assembly file as below.

```
start:  jp   mstart
        ;
        ; data area
        ; including variables,screen & sprite patterns
        ; and sound data
        ;
mstart: di
        ; do initialization
        ; and setup screens
        ;
mloop:  ; start main loop
        ; where everything happens
        jp   mloop
        ;
        ; subroutine section for main loop routines
        ;

        ;
        ; do system dependent routines
        ;

        ; add 100H to the end of your program and set
        ; the stack to that
        end
```

8. Write all the routines around the game and include in your basic assembly file until it is completed.

9. Debug fully and then test run your little project which by this has probably taken 6 months.

Until next time...

See Ya later

Here is a small program for SVI machines written by D. KEIRS of Victoria and he writes:

It is a fairly simple program that shows how to draw and color a pattern or picture on the graphics screen without having to resort to the slow speed of BASIC.

It hopefully will get others started on the road to writing MCR's as there is so little MCR work that is available for the SPECTRAVIDEO.

```
ADD   LABEL  MNEMONIC      OP-CODE     COMMENT

D300  start  call 3610h    cd 10 36    set up screen 1
d303         ld hl,0100h   21 00 01    pattern start address
d306         call 373ch    cd 3c 37    set write mode
d309         ld c,30h      0e 30       block count
d30b  loop2  ld hl,d000h   21 00 d0    data start address
d30e         ld b,10h      06 10       byte count
d310  loop1  ld a,(hl)     7e          get byte
d311         out (80h),a   d3 80       send byte to vram
d313         inc hl        23          next byte
d314         dec b         05          byte count -1
d315         jr nz,loop1   20 f9       if not last byte then next
d317         dec c         0d          block -1
d318         jr nz,loop2   20 f1       if not last block then next
d31a         ld hl,2100h   21 00 21    color table start
d31d         call 373ch    cd 3c 37    set write mode
d320         ld bc,02ffh   01 ff 02    byte count
d323  loop3  ld a,13h      3e 13       color code(black on    )
d325         out (80h),a   d3 80       send color
d327         dec bc        0b          byte count -1
d328         ld a,b        78
d329         or c          b1          last byte?
d32a         jr nz,loop3   20 f7       if not then next
d32c  dly    call 403dh    cd 3d 40    get keypress
d32f         cp 41h        fe 41       is it 'A'
d331         jr nz,dly     20 f9       if not then wait
d333         call 3541h    cd 41 35    screen 1
d336         ret           c9          return to basic
```

The above routine sends a 16 byte data block to screen 1 to create a brick wall pattern. This block is repeated 48 times giving a pattern across half of the top third of the screen. The routine can be easily altered to send more complex patterns as long as the relationship between pattern block and color block is worked correctly.
The more complex the pattern the larger the area of memory needed to store the data, therefore if any section repeats itself the above loop arrangement saves a lot of valuable space. Otherwise you would need a block of 2ffh bytes to store the same size pattern.

I hope this will be of some interest and I hope to see someone writing in with a more complex version.
As you can see when you run this that it is very much faster than poor old basic.

Please note that the above routine is written for a tape based system. If you are using a disk system then the routine should be moved down to a300h with a data file at a000h.
I have not given my data file but I am sure that those who use this will want to experiment with different sets of data.
Any symmetrical pattern will give a nice display.

CP/M

Here is a small program, sent in by Ross Moore, for CP/M users to enter in.

CP/M Colour Change. If you want to use CP/M (40-column) on a green (or any other colour) screen, or simply want to cut your CP/M teeth on something very simple, try the following little program.

On the prompt A> enter the following:-

```
A>  DDT                  <CR>
-     A100               <CR>
0100  MVI  A,C1          <CR>
0102  OUT  81            <CR>
0104  MVI  A,87          <CR>
0106  OUT  81            <CR>
0108  RET                <CR>
0109  <CR>
-     ^C
A> SAVE 1 GREEN.COM  <CR>
```

Then, whenever you type GREEN <CR>, with that file on your disk, presto!! A green screen.

The first line of the program uses the value C1 (hex).
C = 12, which is dark green, and 1 is black.
ie. To return to White (15) print on Blue (4) screen, change line 0100 to MVI A,F4 repeat the above procedure, and:-

```
A> SAVE 1 BLUE.COM <CR>
```

likewise, for Cyan (7) on Black (1) use the value 71 in line 0100 . You can use any colour combination that takes your fancy. Experiment, and see what combinations look the best.

The following MSX game reviews were suplied by, and are available from :-

ROGER STOCKBURGER
MSX SOFTWARE IMPORTS
BASE RADIO SECTION
RAAF BASE RICHMOND
N.S.W. 2755
PH: (045) 78 4972 after 6pm EST.

****
* **ROLLERBALL** *
****

Written By: HAL LABORATORY

The ultimate PINBALL game! I have never been impressed by anything less than the PINBALL CONSTRUCTION SET. This game is EXCELLENT!!

ROLLERBALL $A49.50 Posted
(Cartridge)

ROCKET ROGER $A38.50
(Cassette)

Written By: ALLIGATA SOFWARE

A levels and platforms game with excellent gameplay and sound.
Very addictive.

********** **ALIEN 8** **********

(Cassette) $A42.95

The SMASH-HIT in Japan:
The ultimate in 3-D action games!
Available now on Cassette.
Disk version coming soon.
The 3-D graphics will AMAZE you!!!!

::::::::::::::::::::::::::::::::::
: **ARKANOID** :
::::::::::::::::::::::::::::::::::

Written by: TAITO (JAPAN)
MSX Conversion: IMAGINE

Unfortunately this game is only available on Cassette. Being an Arkanoid addict,I loaded this one up first and after a few minutes of loading: WOW!!!

To sum this up, Arkanoid is...
SHEER BRILLIANCE!
By far the best game I have ever seen on MSX anywhere!! The graphics and sound can only be described as superb and the conversion from the Arcade game is excellent.

My only criticism is that the game is controlled by a Joystick instead of a Paddle, as in the Arcade game. Nevertheless, playability is very good even though the action can get very fast in the higher stages.

The ultimate breakout-type of game, this one I recommend anyone to add too their MSX software collection.

ARKANOID $A49.95
(Cassette)

FINALLY, It's available in Australia:

****
AACKOSOFT'S **FLIGHT DECK**
****

Featuring ** SPEECH **
Full review soon.
Available on Cassette: $A38.50

REVIEW... CX VOICE MASTER

ATTENTION ALL CX5M OWNERS! At last, here is a program just for you! The CX VOICE MASTER is a disk or tape based voice editor for all versions of the CX5M Music Computer, and even at first glance you can't help but be impressed with the ease and versatility of this nifty utility.

Using the Voicing Program YRM-101 (or its update, the YRM-501) it is possible to create files of 48 user-programmed voices (sounds), which can be played either in real-time ("live") or in compositions written using the FM MUSIC COMPOSER (YRM-101 or YRM-501). This is great, but so far there has been no internal way to compile these voices into other combinations. For instance, if you have a great Brass sound on file A, and a nice Piano sound on file B, you can't use both sounds in the same composition... but that's where the CX VOICE MASTER comes in.

The CX VOICE MASTER is available in disk or tape format, but as the copy supplied for review was the tape version, I will deal with the latter. To load, all you do is type "RUN VM" (or, if a disk drive is connected, "RUN"CAS:VM") and wait 2 minutes 45 seconds while a prompt tells you that the program is loading. When ready, the title screen comes up, complete with confirmation of which version of the sound generator you have in your machine, and whether a disk drive is installed (the tape version recognizes disk drives and will happily work with disk files as well as tape files... this is referred to in the manual as a "SMART-TAPE"). Next comes the command prompt, >, just like MSX-DOS and CP/M.

Commands are two characters long, (although when they are abbreviations of English words the full word may also be typed in), followed in most cases by a space, then an argument. The commands are all self-explanatory, but if you happen to forget, a help page is always nearby (just type "HE" or "HELP"). Error messages also abound, to guide the fumble-fingered typist (me) or the brash "i'll bluff my way through without reading the manual" user (also me!) from wreaking havoc with precious only copies (TUT TUT!) of files.

HOW IT WORKS

First you select disk-based or cassettte- based input/output (default is disk if you have a drive connected). Now you are ready to load an input file, i.e. a file from which you want to select some voices. A COPY command then enables you to copy any of the voices from the input file to the output file (the new one you want to create). This is done one voice at a time, and you can actually choose the position in the new file as you go. For example, you can take voice no. 3 from the input file and copy it into position 17 of the output file... a very handy feature! There is even a command which lets you rename the voices in the output file if you want (though this would make it hard to keep track of what you're doing).

When you've taken your pick from the input file, simply load in another one and copy what you want from that one, and so on until you've filled up the output file (48 voices) or you've had enough. The output file can then be saved (on disk or tape) with its own filename. At any time, a display of what's in either the input or the output file can be called up, and there is even a feature which sends you to the "CALL MUSIC" function of the CX5 to play any of the sounds in either input or output file without destroying your copying procedure (just hit [ESC] to get back to where you left off). The usual disk commands FILES and KILL (called CLEAR here) are available, and of course the HELP page.

There is even a totally unexpected bonus for disk users. There are several extra commands which can used to save and load MIDI (Musical Instrument Digital Interface) data to disk, such as external synthesiser patches, drum unit data, etc. Unfortunately, I didn't have any external MIDI equipment on hand at the time of this review to test this feature, however the facility does exist.

Owners of the tape-only SFG-01 sound module (the original one supplied in the CX5 MARK 1) will be delighted to know that this program allows voices to be stored on disk (as long as this program is around to load them back in again).

My only complaints in testing the CX VOICE MASTER were that copying many voices is a bit slow, as the COPY command must be typed in each time, and that the output voice display isn't updated as you copy across new voices (you have to call up the output voice list each time you want to see what has changed. Also, with a voice list on display, there is only room for 3 commands before the top of the list gets scrolled up... better would have been to write new commands over the top of the previous ones to prevent scrolling. However, considering the overall scope and flexibility of the program (even to the point of not needing an update if you upgrade from tape to disk) these are minor criticisms indeed! In fact, William Montgomery, the author, deserves a resounding cheer for addressing the weakest point in the Yamaha CX5M system.

Reviewed by:- **Erik Kowarski**

This program was supplied by...

MSX COMPUTERS & SOFTWARE
11 BRADPOLE ROAD,
ELIZABETH WEST,
S.A. 5113

The prices are :-

CASSETTE $79.95 DISC $86.95 p/p $2.00

DEALER ENQUIRY INVITED

This versatile, easy to use program will allow you to create your own Word-Search puzzles, where words are hidden within a square of nonsense letters, and you then search for the hidden letters.
The entire program is menu-driven and is therefor easy to use.
Please make sure you have some sort of printer connected, one that will print 75 Collumns or more, or else your word-square size will be restricted to the maximum number of columns of the Printer

```
10 ' /-----------------------------------
\
20 ' |  WORDSQUARE COMPILER (MSX/MSX2)
|
30 ' |
|
40 ' |  ROGER STOCKBURGER MSX SOFTWARE
|
50 ' |        WRITTEN 15/11/1986
|
60 ' |
|
70 ' \-----------------------------------
/
80 '
90 '
100 '
110 SCREEN 0:WIDTH 40:KEYOFF:COLOR 15,1
,1:CLS
120 PRINT"/-----------------------------
----------\|W O R D S Q U A R E    C O
M P I L E R|\-----------------------------
------------/";
130 PRINT" [C]1985 ROGER STOCKBURGER MS
X SOFTWARE "
140 PRINT"This versatile, easy to use p
rogram willallow you to create your own
 Word-Searchpuzzles, where words are hi
dden within asquare of nonsense letters
, and you thensearch for the hidden let
ters."
150 PRINT"The entire program is menu-dr
iven and istherefor easy to use."
160 PRINT"Please make sure you have so
me sort ofprinter connected, one tha
t will print75 Collumns or more, or el
se your word-square size will be rest
ricted to themaximum number of columns
 of the Printer"
170 PRINT:PRINT"----------------------
------------------| PRESS <RETURN> TO ST
ART THE PROGRAM: |--------------------
--------------------"
180 IF INKEY$<>"" THEN 180
190 IF INKEY$<>CHR$(13) THEN 190
200 CLS
210 CLEAR 8000,55000!:DIM FF$(80),S$(80
),W$(80),CC(80),RR(80),L(80),E$(2,2)
220 MC=76
230 FOR I=-1 TO 1
240 FOR J=-1 TO 1
250 READ E$(I+1,J+1)
260 NEXT J
270 NEXT I
280 DATA"NW"," N","NE"," W","  "," E","
SW"," S","SE"
290 FOR I=1 TO MC
300 G$=G$+" "
310 NEXT I
320 FOR I=1 TO 8
330 READ D(1,I),D(2,I)
340 NEXT I
350 DATA -1,-1,-1,0,-1,1,0,-1
360 DATA 0,1,1,-1,1,0,1,1
370 GOTO 1350
380 ' * SHELL SORT *
390 PRINT"* SORTING - PLEASE WAIT *"
400 X=1
410 X=2*X
420 IF X<=W0 THEN 410
430 X=INT(X/2)
440 IF X<>0 THEN 460
450 RETURN
460 FOR Y=1 TO W0-X
470 Z=Y
480 A=Z+X
490 IF L(Z)>=L(A) THEN 580
500 X$=W$(Z)
510 W$(Z)=W$(A)
520 W$(A)=X$
530 B=L(Z)
540 L(Z)=L(A)
550 L(A)=B
560 Z=Z-X
570 IF Z>0 THEN 480
580 NEXT Y
590 GOTO 430
600 ' * HIDE WORDS *
610 FOR X=1 TO W0
620 FOR Y=1 TO 50
630 R1=INT(RND(1)*R0)
640 C1=INT(RND(1)*C0)
650 D1=INT(RND(1)*8)+1
660 O1=D1
670 DX=D(1,D1)
680 DY=D(2,D1)
690 IF R1+DX*L(X)<1 OR R1+DX*L(X)>R0 OR
 C1+DY*L(X)<1 THEN 710
700 IF C1+DY*L(X)<+C0 THEN 750
710 D1=D1*(D1<8)*(1=1)+1
720 IF D1<>O1 THEN 670
730 NEXT Y
740 GOTO 920
750 FOR Z=1 TO L(X)
760 IF MID$(W$(X),Z,1)<"A" OR MID$(W$(X
),Z,1)>"Z" THEN 800
770 R1=R1+DX
780 C1=C1+DY
790 IF MID$(S$(R1),C1,1)<>" " AND MID$(
S$(R1),C1,1)<>MID$(W$(X),Z,1) THEN 710
800 NEXT Z
810 FOR Z=L(X) TO 1 STEP -1
820 IF MID$(W$(X),Z,1)<"A" OR MID$(W$(X
),Z,1)>"Z" THEN 890
830 S$(R1)=MID$(S$(R1),1,C1-1)+MID$(W$(
X),Z,1)+MID$(S$(R1),C1+1)
840 RR(X)=R1
850 CC(X)=C1
```

```
860 FF$(X)=E$(DX+1,DY+1)
870 R1=R1-DX
880 C1=C1-DY
890 NEXT Z
900 NEXT X
910 GOTO 1020
920 GOSUB 1850
930 PRINT"SORRY, I AM UNABLE TO FIT WOR
D NUMBER":PRINTSTR$(X);" , ";W$(X)
940 PRINT"INTO THE GRID.":PRINT"OPTIONS
: [1] - SKIP":PRINT"          [2] - STAR
T OVER":PRINT"          [3] - TRY AGAIN"
950 A$=INKEY$:IF A$="" THEN 950
960 IF A$="2" THEN 1790
970 IF A$="3" THEN 620
980 IF A$="1" THEN 1000
990 GOTO 950
1000 W$(X)="/"
1010 GOTO 900
1020 FOR X=1 TO R0
1030 FOR Y=1 TO C0
1040 IF MID$(S$(X),Y,1)<>" " THEN 1060
1050 S$(X)=MID$(S$(X),1,Y-1)+CHR$(INT(2
6*RND(1)+65))+MID$(S$(X),Y+1)
1060 NEXT Y
1070 NEXT X
1080 ' * COMPLETED *
1090 PRINT
1100 PRINT"OPERATION COMPLETE. ENTER FI
LE NAME:"
1110 INPUT T$
1120 SL=0
1130 PRINT
1140 PRINT"DO YOU WANT TO PRINT THE SOL
UTION [Y/N]"
1150 GOSUB 1310
1160 IF A$="N" OR A$="n" THEN 1180
1170 SL=1
1180 GOSUB 1930
1190 GOSUB 1850
1200 F=0
1210 PRINT"DO YOU WANT TO COMPILE ANOTH
ER GRID?    PRESS [Y]/[N]"
1220 GOSUB 1310
1230 IF A$="Y" OR A$="y" THEN 1250
1240 CLS:END
1250 PRINT
1260 PRINT"DO YOU WANT TO USE THE SAME
WORDS?      PRESS [Y]/[N]"
1270 GOSUB 1310
1280 IF A$="N" OR A$="n" THEN 1410
1290 F=1
1300 GOTO 1470
1310 A$=INKEY$:IF A$="" THEN 1310
1320 IF A$<>"Y" AND A$<>"y" AND A$<>"N"
 AND A$<>"n" THEN 1310
1330 RETURN
1340 ' * INIT *
1350 GOSUB 1860
1360 LL=6
1370 GOSUB 1870
1380 PRINT"          WORDSQUARE"
1390 LL=4
1400 GOSUB 1870
1410 FOR I=1 TO W0
1420 W$(I)=""
1430 L(I)=0
1440 NEXT I
1450 PRINT"> ENTER THE NUMBER OF WORDS
TO BE USED    IN THIS WORDSQUARE:"
1460 INPUT W0
1470 PRINT
1480 PRINT"> ENTER ROWS AND COLLUMNS IN
 THE GRID:    (MAX. = 75 FOR STANDARD P
RINTER)"
1490 INPUT R0,C0
1500 PRINT
1510 PRINT
1520 IF R0*C0>+10*W0 THEN 1570
1530 PRINT"I AM UNABLE TO DO THIS. PLEA
SE RE-ENTER:"
1540 FOR I=1 TO 1000
1550 NEXT I
1560 GOTO 1470
1570 PRINT"> I CAN DO THIS WORDSQUARE..
."
1580 IF C0<+MC THEN 1600
1590 PRINT"> BUT IT WILL NOT FIT ON THE
 PAPER..."
1600 IF F=1 THEN 1790
1610 LL=3
1620 GOSUB 1870
1630 PRINT"*  ENTER THE ";STR$(W0);" WO
RDS,":PRINT"TO CORRECT AN ERROR, ENTER
X"
1640 PRINT
1650 FOR I=1 TO W0
1660 PRINT"* WORD NUMBER ";I;":"
1670 INPUT X$
1680 IF LEN(X$)<=R0 AND LEN(X$)<=C0 AND
 X$<>"X" THEN 1740
1690 IF X$<>"X" THEN 1720
1700 I=I-(I>1)*(1=1)
1710 GOTO 1660
1720 PRINT"*-*-* ERROR: THE WORD IS TOO
 LONG: *-*-*"
1730 GOTO 1660
1740 W$(I)=X$
1750 L(I)=LEN(X$)
1760 NEXT I
1770 GOSUB 1860
1780 GOSUB 390
1790 PRINT
1800 PRINT"* PLEASE WAIT, I'M THINKING!
 *"
1810 FOR I=1 TO R0
1820 S$(I)=LEFT$(G$,C0)
1830 NEXT I
1840 GOTO 610
1850 '
1860 RETURN
1870 FOR I=1 TO LL
1880 PRINT
1890 NEXT I
1900 RETURN
1910 GOTO 610
1920 ' * HARD COPY! *
1930 OPEN "LPT:" FOR OUTPUT AS #1
1940 PRINT#1,"*  ROGER STOCKBURGER MSX
SOFTWARE (AUSTRALIA)  *"
1950 PRINT#1,:PRINT#1,"  WORDSQUARE COM
PILER [C]1985 ROGER STOCKBURGER"
```

```
1960 PRINT#1,:PRINT#1,:PRINT#1,T$:PRINT
#1,:PRINT#1,
1970 PRINT#1,"    ";:FOR I=1 TO C0:IF I
/10<>INT(I/10) THEN PRINT#1," ";:GOTO 1
990
1980 PRINT#1,MID$(STR$(I),2,1);
1990 NEXT I:PRINT#1,
2000 PRINT#1,"    ";:FOR I=1 TO C0:PRIN
T#1,RIGHT$(STR$(I),1);:NEXT I:PRINT#1,
2010 FOR X=1 TO R0:IF X<10 THEN PRINT#1
," ";
2020 PRINT#1,STR$(X)" ";
2030 FOR Y=1 TO C0:PRINT#1,MID$(S$(X),Y
,1);
2040 NEXT Y:PRINT#1,:NEXT X:PRINT#1,:PR
INT#1,:PRINT#1,"WORD LIST:"
2050 FOR X=1 TO W0:IF W$(X)="/" THEN 20
70
2060 PRINT#1,W$(X)
2070 NEXT X:FOR I=1 TO 5:PRINT#1,:NEXT
I:IF SL=0 THEN 2110
2080 PRINT#1,"SOLUTION LIST:":PRINT#1,"
WORD                ROW   COLUMN   D
IR":FOR X=1 TO W0:IF W$(X)="/" THEN 210
0
2090 PRINT#1,W$(X);LEFT$(G$,25-LEN(W$(X
)));RR(X);LEFT$(G$,8-LEN(STR$(RR(X))));
CC(X);LEFT$(G$,6-LEN(STR$(CC(X))));FF$(
X)
2100 NEXT X
2110 CLOSE #1:ON ERROR GOTO 0:RETURN
2120 CLOSE #1:PRINT" *-*-* PRINTER ERRO
R - TRY AGAIN!  *-*-*"
2130 PRINT:PRINT"PRESS <RETURN> TO CONT
INUE..."
2140 IF INKEY$<>"" THEN 2140
2150 RESUME 1950
```

USER GROUPS

SYDNEY

We have been advised of a group in Sydney for Spectravideo owners-
Please contact Malcolm Samuels on 888 7664 after hours.

QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mrs Lucille Parker(Secretary)
25 Primrose Street,Woodridge,Queensland 4114

VICTORIA

MELBOURNE SPECTRAVIDEO & MSX USERS GROUP
C/- John Halford
48 Shawlands Avenue,Blackburn South Victoria 3130
Phone (03)878 6169

GIPPSLAND MSX USER GROUP
C/- George Francis
31 Donald Street,Morwell Victoria 3840 ---- Phone (051)34 5175

WESTERN AUSTRALIA

THE MSX USERS GROUP OF W.A
C/- Max Cowin
PO Box 87 Mt Hawthorn W.A. 6016 ---- Phone (09)417 3323

NEW ZEALAND

WELLINGTON SPECTRAVIDEO USERS GROUP
PO Box 26-050 Newlands,Wellington 6004, New Zealand

CANADA

SMUG of Montrel
20 - 159th Avenue N.D. Ile Perrot, Quebec J7V 5V6 Canada

FINLAND

SVI/MSX Club/Turku
C/- Mika Ristolainen
Lumikonkatu 7H 272 SF-20240 Turka Finland

AIR CHOPPER
by Craig Hursey

This listing is in Spectravideo SV318/328 format.

```
10 COLOR 15,1,1
20 GOTO 50
30 PUT SPRITE 5,(36,183),15,3:LOCATE40,
184::COLOR15:PRINT" Copyright Craig Hur
sey 1986"
40 RETURN
50 REM ************************
60 REM **                      **
70 REM **  Approx 6.6k long    **
80 REM **                      **
90 REM **    AIRCHOPPER        **
100 REM**    iiiiiiiiii        **
110 REM**       By             **
120 REM**                      **
130 REM**  (c) Copyright       **
140 REM** Craig Hursey 1986     **
150 REM**                      **
160 REM**       Version 6      **
170 REM**                      **
180 REM**   Date:2/12/86       **
190 REM**                      **
200 REM************************
210 SCREEN1,2
220 FORT=1TO8:READA:C$=C$+CHR$(A):NEXTT
230 SPRITE$(3)=C$
240 DATA 60,66,157,161,161,157,66,60
250 GOSUB 650
260 COLOR1:LOCATE80,20:PRINT"AIRCHOPPER
"
270 GOSUB 30
280 COLOR1:LOCATE80,19:PRINT"AIRCHOPPER
"
290 LOCATE80,29:PRINT"iiiiiiiiii"
300 PRINT
310 COLOR15:PRINT"This game of AIRCHOPP
ER has 7 screens."
320 PRINT
330 PRINT"This game requires a JOYSTICK
 to play."
340 PRINT:
350 PRINT"Your Helicopter has a Turbo b
y which "
360 PRINT
370 PRINT"you use the SPACEBAR ."
380 PRINT
390 PRINT"You use the joystick for ▪ &
 ▪ & ▪ "
400 PRINT
410 PRINT"And the FIRE BUTTON for upwar
ds thrust."
420 PRINT:
430 PRINT"You can't crash into anything
 BLACK!"
440 PRINT
450 PRINT"You fly AIRCHOPPER through th
e ORANGE HOLES"
460 COLOR1:PRINT"          Press any key
to start.":
470 FORX=0TO256:PUTSPRITE1,(X,170),1:NE
XTX
480 A$=INKEY$:IFA$=""THEN GOTO470
490 GOSUB1630:GOTO 690
500 CLS
510 COLOR 15,4,4
520 LINE(0,140)-(100,140),1
530 LINE-(120,50),1
540 LINE-(127,150),1
550 LINE-(145,150),1
560 LINE-(154,50),1
570 LINE-(170,140),1
580 LINE-(256,140),1
590 PAINT(126,160),1
600 LINE(128,150)-(143,145),9,BF
610 X=0:Y=10:LOCATE0,0:COLOR15
620 LINE(0,0)-(256,10),5,BF
630 GOSUB 30
640 RETURN
650 FORT=1TO32:READA:S$=S$+CHR$(A):
660 NEXTT
670 SPRITE$(0)=S$:S$="":A=0:FORT=1TO32:
READA:S$=S$+CHR$(A):NEXTT:SPRITE$(1)=S$
680 RETURN
690 REM MAIN LOOP
700 GOSUB1990
710 LINE(0,0)-(256,10),1,BF
720 SC=0:LE=1:L=3:SP=1
730 LOCATE1,1:COLOR15:PRINT" LEVEL:";LE
;" SCORE:";SC;"LIVES:";L;"SKILL:";ST/.2
5
740 LOCATE1,2:COLOR15:PRINT" LEVEL:";LE
;" SCORE:";SC;"LIVES:";L;"SKILL:";ST/.2
5
750 IF STRIG(0) AND SP=1 THEN X=X+10:HS
=1 ELSE IF STRIG(0) AND SP=0 THEN X=X-1
0:HS=1
760 IF STRIG(1) THEN SP=SP:Y=Y-3
770 D=STICK(1)
780 IF D=3 THEN X=X+2:Y=Y:SP=1
790 IF D=5 THEN Y=Y+2:X=X:SP=SP
800 IF D=7 THEN X=X-2:Y=Y:SP=0
810 IF SP=1 THEN PUT SPRITE0,(-16,-16),
0 ELSE IF SP=0 THEN PUT SPRITE1,(-16,-1
6),0
820 PUT SPRITESP,(X,Y),7
830 IF POINT(X+16,Y+9)=9 OR POINT(X,Y+1
4)=9THEN GOSUB 1050 ELSE IF POINT(X+16,
Y+10)=1 THEN GOSUB 960
840 IF POINT(X,Y+6)=1 THEN GOSUB 960
850 IF POINT(X+8,Y+16)=1 THEN GOSUB 960
860 IF X=<0 THEN X=0 ELSE IF X=>242 THE
NX=242
870 IF Y=<0 THEN Y=0 ELSE IF Y=>178 THE
NY=178
880 Y=Y+1:X=X+ST
890 SOUND6,37
900 SOUND7,35
910 SOUND8,19
920 SOUND12,HS
930 SOUND13,12
940 HS=4
950 GOTO 750
960 REM LOSE LIFE ROUTINES
970 L=L-1:IF L=0 THEN GOSUB 1410
980 X=X :Y=Y:FORT=0TO3:PUTSPRITET,(-16,
-16),0:NEXTT:
```

```
990 ON LE GOSUB1990,1140,1330,1910,2440
.2580,500
1000 SC=LE*100
1010 LINE(0,0)-(256,8),1,BF:COLOR15:LOC
ATE 0,0::PRINT" LEVEL:";LE:"SCORE:";SC:
"LIVES:";L:"SKILL:";ST/.25
1020 LOCATE 0,1:PRINT" LEVEL:";LE:"SCOR
E:";SC;"LIVES:";L;"SKILL:";ST/.25
1030 GOSUB 30
1040 RETURN
1050 REM LANDING PAD
1060 X=X :Y=Y:FORT=0TO3:PUTSPRITET,(-16
,-16),0:NEXTT:
1070 LE=LE+1:IF LE=8 THEN GOTO 2660
1080 ON LE GOSUB1990,1140,1330,1910,244
0,2580,500
1090 SC=LE*100
1100 LINE(0,0)-(256,8),1,BF:COLOR15:LOC
ATE 0,0:PRINT" LEVEL:";LE:"SCORE:";SC:"
LIVES:";L;"SKILL:";ST/.25
1110 COLOR15:LOCATE 0,1:PRINT" LEVEL:";
LE:"SCORE:";SC;"LIVES:";L;"SKILL:";ST/.
25
1120 GOSUB 30
1130 RETURN
1140 REM SCREEN 2
1150 LINE(0,0)-(256,192),1,BF
1160 X=5:Y=120
1170 LINE(0,120)-(45,120),4
1180 LINE-(45,50),4
1190 LINE-(180,50),4
1200 LINE-(170,140),4
1210 LINE-(256,156),4
1220 LINE(0,140)-(75,140),4
1230 LINE-(75,70),4
1240 LINE-(155,70),4
1250 LINE-(140,150),4
1260 LINE-(256,192),4
1270 PAINT(20,125),4:
1280 PUT SPRITE1,(X,Y),8
1290 LINE(256,192)-(253,178),9,BF
1300 LOCATE10,10:COLOR15:PRINT"Press an
y key to continue"
1310 A$=INKEY$:IFA$=""THENGOTO1310 ELSE
 LINE(10,10)-(256,20),1,BF
1320 RETURN
1330 REM screen 3
1340 CLS:LINE(0,0)-(256,192),1,BF
1350 DRAW"BM0,30;C5F50D10R30E10U8E15R10
F30D23F10E20R15F20D25F20R50"
1360 DRAW"BM0,70;C5F50R20D10E25U20E5F10
D30F20D5F10R3E20R15D10F20R5D5R70"
1370 PAINT(0,45),5
1380 LINE(256,165)-(253,185),9,BF
1390 X=0:Y=40:
1400 RETURN
1410 COLOR15:CLS
1420 LOCATE80,50:PRINT"    AIRCHOPPER"
1430 LOCATE80,51:PRINT"    AIRCHOPPER"
1440 PRINT:
1450 PRINT:
1460 SC=LE*100
1470 LOCATE70,100:PRINT"SCORE:";SC
1480 LOCATE70,99:PRINT"SCORE:";SC
1490 PRINT
1500 PRINT
1510 LOCATE70,120:PRINT"LEVEL:";LE
1520 LOCATE70,121:PRINT"LEVEL:";LE
1530 PRINT:
1540 PRINT:
1550 LOCATE70,140:PRINT"SKILL:";ST/.25
1560 LOCATE70,141:PRINT"SKILL:";ST/.25
1570 PRINT:
1580 PRINT:
1590 LOCATE70,180:PRINT" ANOTHER GAME (
Y/N) ?"
1600 LOCATE70,181:PRINT" ANOTHER GAME (
Y/N) ?"
1610 A$=INKEY$:IF A$=""THENGOTO1610 ELS
E IF A$="N"OR A$="n" THEN SCREEN0,1:CLS
:ENDELSE IF A$="Y" OR A$="y" THEN RUN
1620 GOTO 1610
1630 CLS
'640 LOCATE 85,90:PRINT" AIRCHOPPER"
1650 LOCATE 85,91:PRINT" AIRCHOPPER"
1660 PRINT
1670 PRINT
1680 PRINT
1690 LOCATE 85,100:PRINT" iiiiiiiiii"
1700 LOCATE 85,99:PRINT" iiiiiiiiii"
1710 PRINT
1720 PRINT
1730 PRINT
1740 LOCATE 70,130: PRINT"SKILL LEVEL:
(1-8) "
1750 LOCATE 70,131: PRINT"SKILL LEVEL:
(1-8) "
1760 GOSUB 30
1770 A$=INKEY$
1780 IF A$=""THEN GOTO 1770
1790 IF A$>"8"ORA$<"1" THEN GOTO 1770
1800 IF A$="1"THEN ST=1*.25
1810 IF A$="2"THEN ST=2*.25
1820 IF A$="3"THEN ST=3*.25
1830 IF A$="4"THEN ST=4*.25
1840 IF A$="5"THEN ST=5*.25
1850 IF A$="6"THEN ST=6*.25
1860 IF A$="7"THEN ST=7*.25
1870 IF A$="8"THEN ST=8*.25
1880 RETURN
1890 DATA 0,0,0,0,0,0,0,255,4,30,47,79,25
5,62,0,0,0,0,0,0,0,0,0,0,248,3,7,254,192,
0,0,0,0,0,0,0
1900 DATA 0,0,0,0,0,0,0,31,192,224,127,3,
0,0,0,0,0,0,0,0,0,0,0,0,0,255,32,120,244,24
2,255,124,0,0,0
1910 REM screen 4
1920 CLS:LINE(0,0)-(256,192),1,BF
1930 DRAW"C5;BM0,50;R15U10e25R60F25D451
12g50R260"
1940 DRAW"C5;BM0,70;R34U10e25R30f10D25g
75D20R260"
1950 PAINT(0,55),5
1960 LINE(250,150)-(256,160),9,BF
1970 X=10:Y=50:
1980 RETURN
1990 REM SCREEN5
2000 CLS
2010 CIRCLE(200,140),30,15,,,1/4
2020 PAINT(200,140),15 :C=2
2030 LINE(200,147)-(200,133),13
2040 LINE(180,140)-(220,140),13
2050 COLOR1
2060 LOCATE 0,50
2070 CIRCLE(30,58),18,1,6.28,3.14
2080 PRINT"  ½½½½½½"
2090 PAINT(30,50),1
2100 PRINT" ☼       ¶"
```

```
2110 PRINT" »»»»»»»»"
2120 PRINT" ¶½½½½½½◇"
2130 PRINT"    IIIIII"
2140 PRINT"    IIIIII"
2150 PRINT"    IIIIII"
2160 PRINT"    IIIIII"
2170 PRINT"    !!!!!i"
2180 PRINT"    IIIIII"
2190 PRINT"    IIIIII"
2200 PRINT"    IIIIII"
2210 PRINT"    IIIIII"
2220 LINE(0,192)-(256,150),12,BF
2230 X=190:Y=118
2240 LINE(58,80)-(175,150),1,BF
2250 LINE(58,80)-(63,75),1
2260 LINE-(170,75),1
2270 LINE-(175,80),1
2280 LINE(58,80)-(175,80),1
2290 PAINT(100,78),1:COLOR9
2300 LOCATE85,90:PRINT"AIRCHOPPER"
2310 LOCATE85,91:PRINT"AIRCHOPPER"
2320 LOCATE110,110:PRINT"By"
2330 LOCATE110,111:PRINT"By"
2340 GOSUB 30:COLOR9: LOCATE80,130:PRIN
T"CRAIG HURSEY"
2350 LOCATE80,131:PRINT"CRAIG HURSEY"
2360 LINE(253,100)-(256,120),9,BF
2370 LOCATE 10,20:COLOR1:PRINT"     Pres
s any key to begin"
2380 A$=INKEY$:IFA$=""THEN GOTO2380ELSE
 LINE(10,20)-(250,30),4,BF
2390 PUT SPRITE 1,(X,Y),7
2400 FOR H=10TO2 STEP-.05
2410 SOUND6,37:SOUND7,35:SOUND8,19:SOUN
D12,H:SOUND13,12:
2420 NEXTH
2430 RETURN
2440 REM SCREEN6
2450 CLS
2460 FORX=0TO 256
2470 LINE(X,0)-(X,INT(RND(1)*103)+1),1
2480 NEXTX
2490 LINE(0,192)-(0,192),1
2500 FORX=0TO240 STEP 5
2510 LINE-(X,182-INT(RND(1)*71+1)),1
2520 NEXTX
2530 LINE-(242,192),1:PAINT(200,190),1
2540 LINE(240,192)-(256,160),4,BF
2550 LINE(252,95)-(256,110),9,BF
2560 X=0 :Y=95
2570 RETURN
2580 REM SCREEN 7
2590 CLS:LINE(0,192)-(0,192),1
2600 FORX=0TO256STEP5:Y=190-INT(RND(1)*
40+8):LINE-(X,Y),1:NEXTX:LINE-(256,192)
,1:PAINT(126,190),1
2610 FOR T=1TO75
2620 X=INT(RND(1)*220)+16:Y=INT(RND(1)*
170)+1:LOCATE X,Y:COLOR1:PRINT"*":NEXTT
2630 X=0:Y=10
2640 :LINE(230,160)-(256,100),4,BF: LIN
E(230,160)-(256,165),9,BF
2650 RETURN
2660 REM END OF GAME
2670 CLS
2680 GOSUB 30
2690 LOCATE70,20:PRINT"    AIRCHOPPER"
2700 LOCATE70,21:PRINT"    AIRCHOPPER"
2710 LOCATE50,29:PRINT"    iiiiiiiiii"
2720 LOCATE50,28:PRINT"    iiiiiiiiii"
2730 PRINT:
2740 PRINT:
2750 :LOCATE50,60:PRINT"    You have los
t";3-L:"lives"
2760 :LOCATE50,61:PRINT"    You have los
t";3-L:"lives"
2770 PRINT:
2780 PRINT:
2790 IF L=3 THEN LOCATE50,100:PRINT"You
 are an ELITE Pilot":LOCATE50,101:PRINT
"You are an ELITE Pilot" ELSE IF L=2 TH
ENLOCATE16,100: PRINT"You are now Eliga
ble for Air Rescue.":LOCATE16,101:PRINT
"You are now Eligable for Air Rescue."
2800 IF L=1 THENLOCATE30,100:PRINT"Sorr
y We don't need you.":LOCATE30,101:PRIN
T"Sorry We don't need you."
2810 PRINT:
2820 PRINT:
2830 SC=LE*100
2840 LOCATE50,120:PRINT"  Score:";SC
2850 LOCATE50,121:PRINT"  Score:";SC
2860 LOCATE50,150:PRINT"  ANOTHER GAME
 (Y/N)"
2870 LOCATE50,151:PRINT"  ANOTHER GAME
 (Y/N)"
2880 A$=INKEY$:IF A$=""THENGOTO2880 ELS
E IF A$="Y" OR A$="y" THEN RUN ELSE IFA
$="N" OR A$="n" THEN CLS:END ELSE GOTO
2880
```

READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

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◇◇◇◇◇ M.S.X. DRAW ◇ DISC BASED ◇◇◇◇◇
                by L. Morellini

M.S.X. DRAW is a program which allows
the actual drawing of graphics direct
to the M.S.X. graphics high res screen
It allows the use of simple key codes
to draw Circles, Lines, Boxes and has
facilities to change color, Edit,Save
Data & Screens, Output to a Printer,
Draw using the cursor keys etc.

This program uses the key pad for
movement. Try moving the cursor and
push the 'L' & 'C' keys. Experiment a
bit to discover the full facilities.

Push the letter 'H' to access the HELP
screen at any time during Normal or
Edit mode.


10 KEY OFF:CLS:SCREEN0:WIDTH39
20 PRINT"◇◇◇◇◇ M.S.X. DRAW ◇ DISC BASED
 ◇◇◇◇◇"
30 LOCATE10,2:PRINT"by L. Morellini":PR
INT
40 PRINT"M.S.X. DRAW is a program which
 allows"
50 PRINT"the actual drawing of graphics
 direct"
60 PRINT"to the M.S.X. graphics high re
s screen"
70 PRINT"It allows the use of simple ke
y codes "
80 PRINT"to draw Circles, Lines, Boxes
and has"
90 PRINT"facilities to change color, Ed
it,Save"
100 PRINT"Data & Screens, Output to a P
rinter,
110 PRINT"Draw using the cursor keys et
c.
120 PRINT:PRINT"This program uses the k
ey pad for"
130 PRINT"movement. Try moving the curs
or and "
140 PRINT"push the 'L' & 'C' keys. Expe
riment a"
150 PRINT"bit to discover the full faci
lities."
160 PRINT:PRINT"Push the letter 'H' to
access the HELP"
170 PRINT"screen at any time during Nor
mal or"
180 PRINT"Edit mode."
190 PRINT:PRINT"  *** PUSH ANY KEY TO P
ROCEED ***"
200 D$=INKEY$:IFD$="" THEN200
210 ONERROR GOTO250
220 MAXFILES=2:KEY OFF:CLEAR2000
230 DEFINT A,I,Y,X,W,V,C,L,S:DEFSTR D,E
,F
240 FORX=1TO10:KEY(X)ON:NEXT:DIMI(400),
K(400),Y(400),X(400),W(400),V(400),C(40
0),L(400):SCREEN2,0,0:C=15:S=10:N=0:COL
ORC,5,4
250 D=CHR$(&B01111100):E=CHR$(&B0100010
0):F=CHR$(&B01010100):SPRITE$(0)=D+E+E+
F+E+E+D:Y=120:X=80:SPRITE$(1)=SPRITE$(0
):PUT SPRITE1,(Y,X),15:W=Y:V=X:GOSUB241
0:GOSUB2420
260 A=STICK(0):D=INKEY$
270 IFA=1THENX=X-S:IFX<10THENX=10
280 IFA=5THENX=X+S:IFX>174THENX=174
290 IFA=3THENY=Y+S:IFY>240THENY=240
300 IFA=7THENY=Y-S:IFY<4THENY=4
310 PUTSPRITE0,(Y,X),1
320 IFD=CHR$(&H18)THENPUTSPRITE1,(Y,X),
15:W=Y:V=X
330 IFD=CHR$(&HB)THENSWAPY,W:SWAPX,V:PU
TSPRITE1,(W,V),15
340 IFD=CHR$(&H7F)THENGOSUB1200:N=N-1:G
OSUB2320:IFN<1THENN=0
350 IFD="l"ORD="L"THENGOSUB470:GOSUB480
360 IFD="b"ORD="B"THENGOSUB470:GOSUB490
370 IFD="f"ORD="F"THENGOSUB470:GOSUB500
380 IFD="c"ORD="C"THENGOSUB470:GOSUB510
390 IFD="w"ORD="W"THENGOSUB2470
400 IFD="p"ORD="P"THENGOSUB470:GOSUB580
410 IFD="d"ORD="D"THENGOSUB1980
420 IFD="e"ORD="E"THENGOSUB890:PUTSPRIT
E1,(W,V),15
430 IFD="m"ORD="M"THENGOSUB2160:SCREEN2
:COLORC,5,4:GOSUB1030:PUTSPRITE1,(W,V),
15
440 IFD="h"ORD="H"THENGOSUB1350:SCREEN2
:COLORC,5,4:GOSUB1030:PUTSPRITE1,(W,V),
15
450 ONKEYGOSUB670,690,1030,620,590,710,
760,1820,850,810
460 GOTO260
470 N=N+1:Y(N)=Y+3:X(N)=X+4:W(N)=W+3:V(
N)=V+4:C(N)=C:GOSUB2320:RETURN
480 LINE(Y+3,X+4)-(W+3,V+4),C:I(N)=1:RE
TURN
490 LINE(Y+3,X+4)-(W+3,V+4),C,B:I(N)=2:
RETURN
500 LINE(Y+3,X+4)-(W+3,V+4),C,BF:I(N)=3
:RETURN
510 IF(W+3)<(Y+3)THENR(1)=(Y+3)-(W+3)
520 IF(Y+3)<(W+3)THENR(1)=(W+3)-(Y+3)
530 IF(Y+3)=(W+3)THENR(1)=ABS((Y+3)-(W+
3))
540 IF(V+4)<(X+4)THENR(2)=(X+4)-(V+4)
550 IF(X+4)<(V+4)THENR(2)=(V+4)-(X+4)
560 IF(X+4)=(V+4)THENR(2)=ABS((X+4)-(V+
4))
570 R=(R(1)^2)+(R(2)^2):R=(SQR(R)):CIRC
LE(W+3,V+4),R,C:I(N)=4:L(N)=R:RETURN
580 I(N)=5:PAINT(Y+3,X+4),C:RETURN
590 C=C+1:IFC>15THENC=1
600 IFC=5THENC=6
610 GOTO640
620 C=C-1:IFC<1THENC=15
630 IFC=5THENC=4
640 OPEN"grp:"FOROUTPUTAS#1:LINE(100,18
2)-(155,191),C,BF:PRESET(110,184),C:IFC
=1ORC=13ORC=4ORC=12ORC=2ORC=6THEN660ELS
E650:COLOR1:PRINT#1,"Color":CLOSE#1:RET
URN
650 COLOR1:PRINT#1,"Color":CLOSE#1:RETU
RN
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660 COLOR15:PRINT#1,"Color":CLOSE#1:RET
URN
670 IFS>19THENRETURN
680 S=S+1:GOSUB2370:RETURN
690 IFS<2THENRETURN
700 S=S-1:GOSUB2370:RETURN
710 SCREEN0:LOCATE9,0:PRINT"** SAVE ROU
TINE **":LOCATE9,2:PRINT"( ENTER 0 TO Q
UIT )":PRINT:PRINT"NOTE ! Enter only NA
ME not file type"
720 LOCATE0,7:PRINT"EXISTING FILES ARE;
-":LOCATE0,9:FILES"*.*":LOCATE9,20:INPU
T"NAME OF FILE; ";J$:IFJ$="0"THEN750
730 OPENJ$+".Dat"FOROUTPUTAS#1:T=N:PRIN
T#1,T
740 FORN=1TOT:PRINT#1,I(N);Y(N);X(N);W(
N);V(N);C(N);L(N);K(N):NEXT:CLOSE
750 SCREEN2:COLORC,5,4:GOSUB1030:PUTSPR
ITE1,(W,V),15:RETURN
760 SCREEN0:LOCATE9,0:PRINT"** LOAD ROU
TINE **":LOCATE9,2:PRINT"( ENTER 0 TO Q
UIT )":PRINT:PRINT"NOTE ! Enter only NA
ME not file type":PRINT"and only load D
AT files from list"
770 LOCATE0,7:PRINT"EXISTING FILES ARE:
-":LOCATE0,9:FILES"*.*":LOCATE9,20:INPU
T"NAME OF FILE; ";J$:IFJ$="0"THEN800
780 OPENJ$+".Dat"FORINPUTAS#1:INPUT#1,T
790 FORN=1TOT:INPUT#1,I(N),Y(N),X(N),W(
N),V(N),C(N),L(N),K(N):NEXT:CLOSE
800 SCREEN2:COLORC,5,4:GOSUB1030:PUTSPR
ITE1,(W,V),15:RETURN
810 SCREEN0:LOCATE5,0:PRINT"** SCREEN S
AVE ROUTINE **":LOCATE9,2:PRINT"(ENTER
0 TO QUIT )":PRINT:PRINT"NOTE ! Only en
ter NAME not file type"
820 LOCATE0,6:PRINT"EXISTING FILES ARE:
-":LOCATE0,8:FILES"*.*":LOCATE9,20:INPU
T"NAME OF FILE; ";J$:IFJ$="0"THEN840
830 SCREEN2:COLORC,5,4:GOSUB1030:BSAVEJ
$+".scr",&H0,&H3FFF,S:PUTSPRITE1,(W,V),
15:RETURN
840 SCREEN2:COLORC,5,4:GOSUB1030:PUTSPR
ITE1,(W,V),15:RETURN
850 SCREEN0:LOCATE5,0:PRINT"** SCREEN L
OAD ROUTINE **":LOCATE9,2:PRINT"(ENTER
0 TO QUIT )":PRINT:PRINT"NOTE ! Only en
ter NAME not file type":PRINT"and only
load SCR files from list"
860 LOCATE0,7:PRINT"EXISTING FILES ARE:
-":LOCATE0,9:FILES"*.*":LOCATE9,20:INPU
T"NAME OF FILE; ";J$:IFJ$="0"THEN880
870 SCREEN2:COLORC,5,4:BLOADJ$+".scr",S
:PUTSPRITE1,(W,V),15:GOSUB2410:GOSUB242
0:RETURN
880 SCREEN2:COLORC,5,4:PUTSPRITE1,(W,V)
,15:GOSUB2410:GOSUB2420:RETURN
890 J=N:N=1:LINE(0,0)-(255,10),3,BF:OPE
N"GRP:"AS1:PRESET(28,2):PRINT#1,"EDIT M
ODE - 'ESC' TO END":CLOSE#1:GOSUB990
900 D=INKEY$
910 IF D="d"ORD="D"THENGOSUB1200:I(N)=0
920 IF D="p"ORD="P"THENGOSUB970
930 IF D="n"ORD="N"THENGOSUB1000
940 IF D=CHR$(27)THENN=J:GOSUB1030:RETU
RN
950 IFD="h"ORD="H"THENGOSUB1690:SCREEN2
:COLORC,5,4:N=J:GOSUB1030:GOTO890
960 GOTO900
970 N=N-1:IFN<1THENN=J
980 IF I(N)=0THEN970
990 PUTSPRITE0,(Y(N)-3,X(N)-4),1:PUTSPR
ITE1,(W(N)-3,V(N)-4),15:RETURN
1000 N=N+1:IFN>JTHENN=1
1010 IF I(N)=0THEN1000
1020 PUTSPRITE0,(Y(N)-3,X(N)-4),1:PUTSP
RITE1,(W(N)-3,V(N)-4),15:RETURN
1030 CLS:T=N
1040 FORN=1TOT
1050 IFI(N)<>1THEN1070
1060 LINE(Y(N),X(N))-(W(N),V(N)),C(N)
1070 IFI(N)<>2THEN1090
1080 LINE(Y(N),X(N))-(W(N),V(N)),C(N),B
1090 IFI(N)<>3THEN1110
1100 LINE(Y(N),X(N))-(W(N),V(N)),C(N),B
F
1110 IFI(N)<>4THEN1130
1120 CIRCLE(W(N),V(N)),L(N),C(N)
1130 IFI(N)<>5THEN1150
1140 PAINT(Y(N),X(N)),C(N)
1150 IFI(N)<>6THEN1170
1160 LINE(Y(N),X(N))-(W(N),V(N)),C(N)
1170 IFI(N)<>7THEN1190
1180 OPEN"GRP:"FOROUTPUTAS#2:PRESET(Y(N
),X(N)),5:COLORC(N):PRINT#2,CHR$(K(N)):
CLOSE#2
1190 NEXT:N=N-1:GOSUB2410:GOSUB2420:RET
URN
1200 IFI(N)<>1THEN1220
1210 LINE(Y(N),X(N))-(W(N),V(N)),5
1220 IFI(N)<>2THEN1240
1230 LINE(Y(N),X(N))-(W(N),V(N)),5,B
1240 IFI(N)<>3THEN1260
1250 LINE(Y(N),X(N))-(W(N),V(N)),5,BF
1260 IFI(N)<>4THEN1280
1270 CIRCLE(W(N),V(N)),L(N),5
1280 IFI(N)<>5THEN1300
1290 PAINT(Y(N),X(N)),5
1300 IFI(N)<>6THEN1320
1310 LINE(Y(N),X(N))-(W(N),V(N)),5
1320 IFI(N)<>7THEN1340
1330 LINE(Y(N),X(N))-(Y(N)+4,X(N)+7),5,
BF
1340 RETURN
1350 SCREEN0:KEY OFF:COLOR1,7
1360 LOCATE3,0:PRINT"** H E L P     S C
R E E N **"
1370 LOCATE,2
1380 PRINT " F 1 - SPEED +       F 2 - S
PEED -"
1390 LOCATE,3
1400 PRINT " F 3 - RE-DRAW       F 4 - P
REV COLOR"
1410 LOCATE,4
1420 PRINT " F 5 - NEXT COLOR    F 6 - S
AVE DATA
1430 LOCATE,5
1440 PRINT " F 7 - LOAD DATA     F 8 - B
LACK IMAGE"
1450 LOCATE,6
1460 PRINT " F 9 - LOAD SCREEN   F10 - S
AVE SCREEN"
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1470 LOCATE,8
1480 PRINT " ■■■■ INDIVIDUAL KEY FUNCTI
ONS ■■■■"
1490 LOCATE,10
1500 PRINT " 'L' - LINE          'B' - B
OX"
1510 LOCATE,11
1520 PRINT " 'F' - BOX FILL      'C' - C
IRCLE"
1530 LOCATE,12
1540 PRINT " 'E' - EDIT MODE     'P' - P
AINT "
1550 LOCATE,13
1560 PRINT " 'D' - DRAW MODE     'W' - W
RITE MODE"
1570 LOCATE,14
1580 PRINT " 'M' - MEMORY LEFT"
1590 LOCATE,16
1600 PRINT " 'DEL'    DELETES LAST ITEM
S DONE"
1610 LOCATE,18
1620 PRINT " 'SELECT' SETS 2nd CURSOR P
OINT"
1630 LOCATE,20
1640 PRINT " 'HOME'   SWAPS CURSORS"
1650 LOCATE,22
1660 PRINT "     ** PRESS 'ESC' TO RETUR
N **"
1670 D=INKEY$:IFD=CHR$(27)THENRETURN
1680 GOTO1670
1690 SCREEN0:KEY OFF:COLOR1,7
1700 LOCATE3,1:PRINT"** H E L P     S C
R E E N **"
1710 LOCATE6,3:PRINT"  E D I T     M O D
 E"
1720 LOCATE,8
1730 PRINT " 'N' - JUMP TO THE NEXT ITE
M"
1740 LOCATE,10
1750 PRINT " 'P' - RETURN TO THE LAST I
TEM"
1760 LOCATE,12
1770 PRINT " 'D' - DELETE ITEM BETWEEN
CURSORS"
1780 LOCATE,22
1790 PRINT "     ** PRESS 'ESC' TO RETUR
N **"
1800 D=INKEY$:IFD=CHR$(27)THENRETURN
1810 GOTO1800
1820 T=N:FORN=1TOT
1830 IFI(N)<>1THEN1850
1840 LINE(Y(N),X(N))-(W(N),V(N)),1
1850 IFI(N)<>2THEN1870
1860 LINE(Y(N),X(N))-(W(N),V(N)),1,B
1870 IFI(N)<>3THEN1890
1880 LINE(Y(N),X(N))-(W(N),V(N)),1,BF
1890 IFI(N)<>4THEN1910
1900 CIRCLE(W(N),V(N)),L(N),1
1910 IFI(N)<>5THEN1930
1920 PAINT(Y(N),X(N)),1
1930 IFI(N)<>6THEN1950
1940 LINE(Y(N),X(N))-(W(N),V(N)),1
1950 IFI(N)<>7THEN1970
1960 OPEN"GRP:"FOROUTPUTAS#2:PRESET(Y(N
),X(N)),1:COLOR1:PRINT#2,CHR$(K(N)):CLO
SE#2
1970 NEXT:RETURN
1980 LINE(0,0)-(255,10),7,BF:PRESET(18,
2),1:OPEN"GRP:"AS1:PRINT#1,"DRAW MODE
-  'ESC' TO END":CLOSE#1:PUTSPRITE1,(0,
0),0
1990 A=STICK(0):D=INKEY$
2000 IFA=1THENX=X-1:IFX<1THENX=1
2010 IFA=8THENX=X-1:Y=Y-1:IFX<1THENX=1
2020 IFA=5THENX=X+1:IFX>182THENX=182
2030 IFA=2THENX=X-1:Y=Y+1:IFX>182THENX=
182
2040 IFA=3THENY=Y+1:IFY>240THENY=240
2050 IFA=4THENY=Y+1:X=X+1:IFY>240THENY=
240
2060 IFA=7THENY=Y-1:IFY<4THENY=4
2070 IFA=6THENY=Y-1:X=X+1:IFY<4THENY=4
2080 IFD=CHR$(&H7F)THENGOSUB1200:N=N-1:
IFN<1THENN=0
2090 IFD=CHR$(27)THENPUTSPRITE1,(W,V),1
5:CLS:GOSUB1030:RETURN
2100 IFA=0THEN1990
2110 PSET(Y+3,X+4),C
2120 PUTSPRITE0,(Y,X),1
2130 IFA<>OTHENGOSUB2150
2140 O=A:J=X+4:K=Y+3:W(N)=K:V(N)=J:GOTO
1990
2150 O=A:N=N+1:I(N)=6:Y(N)=Y+3:X(N)=X+4
:C(N)=C:GOSUB2320:RETURN
2160 SCREEN0:KEY OFF:COLOR1,7
2170 PRINT"** M E M O R Y   R E M A N I
 N G **"
2180 LOCATE4,4
2190 PRINT " FREE USR AREA    :- ";FRE(0
)
2200 LOCATE4,6
2210 PRINT " ITEMS REMAINING :- ";400-N
2220 LOCATE4,8
2230 PRINT " ITEMS SO FAR     :- ";N
2240 LOCATE0,12
2250 PRINT " NOTE ! Only 400 STEPS avai
lable."
2260 LOCATE0,14
2270 PRINT " Once they,re used you're s
tuffed."
2280 LOCATE0,18
2290 PRINT " PRESS ' ESC ' TO RETURN TO
 PROG."
2300 D=INKEY$:IFD=CHR$(27)THENRETURN
2310 GOTO2300
2320 REM
2330 OPEN"grp:"FOROUTPUTAS#1
2340 LINE(60,182)-(100,191),14,BF:COLOR
1
2350 PRESET(58,184),14:PRINT#1,N
2360 CLOSE#1:RETURN
2370 OPEN"grp:"FOROUTPUTAS#1
2380 LINE(214,182)-(255,191),14,BF:COLO
R1
2390 PRESET(219,184),14:PRINT#1,S
2400 CLOSE#1:RETURN
2410 LINE(0,0)-(255,10),11,BF:COLOR1:PR
ESET(52,2),11:OPEN"GRP:"AS1:PRINT#1,"M
S X  -  D R A W":CLOSE#1:RETURN
2420 OPEN"grp:"FOROUTPUTAS#1
2430 LINE(3,182)-(255,191),14,BF:COLOR1
```

```
2440 PRESET(18,184),14:PRINT#1,"Step ";
N;"          Scale ";S
2450 LINE(100,182)-(155,191),C,BF
2460 CLOSE#1:GOSUB640:RETURN
2470 LINE(0,0)-(255,10),9,BF:COLOR1:PRE
SET(28,2),9:OPEN"GRP:"AS1:PRINT#1,"WRIT
E MODE - 'ESC' TO END":CLOSE#1
2480 S=6:GOSUB2370:OPEN"grp:"FOROUTPUTA
S#2
2490 PUTSPRITE1,(Y,X),0:PUTSPRITE1,(Y,X
),0:SCREEN,1:D=CHR$(&B11110000):E=CHR$(
&B00000000):SPRITE$(3)=E+E:E+E+D+E+E+E:
PUT SPRITE3,(Y-1,X-1),15:W=Y:V=X
2500 A=STICK(0):D=INKEY$
2510 IFA=1THENX=X-S:IFX<13THENX=171
2520 IFA=5THENX=X+S:IFX>171THENX=13
2530 IFA=3THENY=Y+S:IFY>240THENY=4:IFX<
164THENX=X+9ELSEX=13
2540 IFA=7THENY=Y-S:IFY<4THENY=240
2550 PUTSPRITE3,(Y,X),15
2560 IFD=CHR$(27)THEN2650
2570 IFD=CHR$(&H7F)THENGOSUB2660:GOSUB2
320:D="":IFN<1THENN=1
2580 IFD=""THEN2500
2590 IFASC(D)=&H20THENY=Y+S
2600 IFASC(D)=&HDTHENX=X+S
2610 IFASC(D)<&H21THEN2500
2620 IFY>240THENY=4:X=X+9:GOTO2550
2630 N=N+1:Y(N)=Y:X(N)=X:C(N)=C:I(N)=7:
K(N)=ASC(D):PRESET(Y,X),5:COLORC:PRINT#
2,D:Y=Y+S:PUTSPRITE3,(Y,X),15:GOSUB2320
2640 GOTO2500
2650 CLOSE#2:SCREEN,0:D=CHR$(&B01111100
):E=CHR$(&B01000100):F=CHR$(&B01010100)
:SPRITE$(0)=D+E+E+F+E+E+D:SPRITE$(1)=SP
RITE$(0):PUT SPRITE1,(Y,X),15:W=Y:V=X:G
OSUB2410:RETURN
2660 LINE(Y(N),X(N))-(Y(N)+4,X(N)+7),5,
BF:PUTSPRITE3,(Y(N),X(N)),15:Y=Y(N):X=X
(N):N=N-1:RETURN
```

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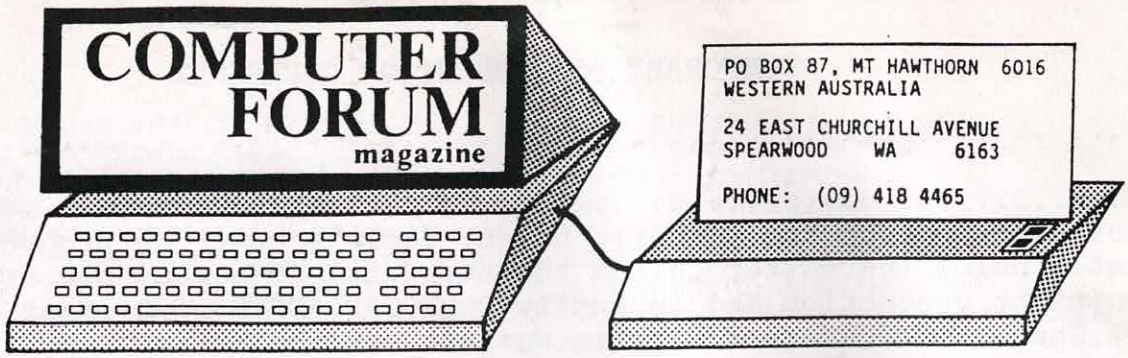
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